Quantum wall dynamics: boundary choice, reflected packets and impulse

Result, 2026-10-27. For the specified incident Gaussian and free Dirichlet half-line dynamics below, the nonnegative boundary-force measures converge weakly on the whole time axis:

$$\frac{\hbar^2}{2m}\,|\partial_q \psi_{\hbar,D}(0,t)|^2\,dt \implies 2|p_0|\,\delta_{t_*}, \qquad t_* = -\frac{mq_0}{p_0} > 0. \tag{1}$$

This includes the impact layer excluded from a uniform off-impact momentum-current estimate. In fact, on the $\sqrt{\hbar}$ time scale the force has an explicit Gaussian profile and the momentum current tends to an error-function transition.

For the corresponding **Neumann** half-line packet, the derivative in (1) vanishes identically. The complete boundary stress has another term; it has the same limiting impulse and the same leading impact-layer profile. The particle's classical bounce therefore does not determine whether the quantum wave satisfies Dirichlet or Neumann boundary conditions. General Robin conditions are discussed through their scattering phases; no wavepacket limit is claimed here for arbitrary Robin parameters.

The boundary contribution to Ehrenfest's theorem is established literature: see De Vincenzo (2013), equations (38)–(39) for a wall on the opposite half-line (passage read), and Albrecht et al. (2023) (abstract) on boundary momentum and bouncing packets. A particularly close precedent is Belloni–Doncheski–Robinett (2004), equations (8)–(11), (16)–(19): they normalize the same odd-image Gaussian on the opposite half-line and analyze its near-collision position and force (relevant passages read). The computation below gives a particular weak impulse measure and compares the Neumann stress; it does not claim priority for the reflected Gaussian or wall forces.

Classical impact and quantum boundary data

Take the half-line $q \geq 0$ and a free particle starting at $q_0 > 0$ with $p_0 < 0$. For an ideal **elastic** wall, the prescribed impact law is $p(t_*+) = -p(t_*-)$ at $t_* = -mq_0/p_0$. The motion is

$$q(t) = |q_0 + p_0 t/m|, \qquad p(t) = p_0 \operatorname{sgn}(q_0 + p_0 t/m) \quad (t \neq t_*).$$

Position is continuous; momentum has a jump. A smooth-force equation with a finite smooth force does not describe that jump. Nonpenetration alone does not select the elastic law: a restitution coefficient or another contact rule would be additional data.

For a smooth wall in several dimensions, the elastic specular rule reverses the normal component and retains the tangential component: $v_+ = v_- - 2(v_- \cdot n)n$. This requires a specified unit normal and a well-defined collision. Corners, grazing contacts and an accumulation of impacts require separate analysis; one should not infer their uniqueness merely from the local smooth-force theorem.

A mathematically defined quantum impenetrable wall can be represented by the Dirichlet Laplacian

$$H_{\hbar,D} = -\frac{\hbar^2}{2m}\frac{d^2}{dq^2}, \qquad D(H_{\hbar,D}) = H^2(0,\infty) \cap H_0^1(0,\infty).$$

It is self-adjoint and gives a unitary group. Its Euclidean kernel is

$$K_D(T;x,y) = k_{\hbar,T}(x-y) - k_{\hbar,T}(x+y), \quad k_{\hbar,T}(r) = \sqrt{\frac{m}{2\pi\hbar T}}e^{-mr^2/(2\hbar T)}, \quad x,y>0.$$

This is the method of images: the Euclidean reference process is Brownian motion killed at the boundary. The quantum real-time wall reflects, but killed Euclidean paths should not

themselves be read as real-time bouncing trajectories. The reflected-packet estimate below proves a quantitative coherent-packet position limit through the bounce.

A subtle domain issue is important: the symmetric differential operator $-i\hbar\, d/dq$ on the half-line with Dirichlet boundary has unequal deficiency indices and no self-adjoint extension on $L^2(0,\infty)$. The Dirichlet Hamiltonian is nevertheless self-adjoint. A current or a quadratic-form expectation involving $-i\hbar d/dq$ can be studied, but it should not be presented as the expectation of a self-adjoint half-line momentum operator. See Al-Hashimi and Wiese, arXiv:2103.01715 (abstract) for the boundary momentum problem.

1. The packet and its boundary value

Fix $m, \omega, q_0 > 0$ and $p_0 < 0$, with $v = p_0/m$. On the full line take the normalized coherent state

$$g_{\hbar,0}(q) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{m\omega(q-q_0)^2}{2\hbar} + \frac{ip_0(q-q_0)}{\hbar}\right].$$

Let $g_{\hbar,t} = e^{-itP^2/(2m\hbar)} g_{\hbar,0}$, $a(t) = q_0 + vt$, $A(t) = 1 + \omega^2 t^2$, and

$$\gamma_\hbar = \langle g_{\hbar,0}, Rg_{\hbar,0}\rangle = \exp\left[-\frac{m\omega q_0^2 + p_0^2/(m\omega)}{\hbar}\right], \qquad (Rg)(q) = g(-q). \tag{2}$$

The odd and even extensions restrict to the normalized half-line states

$$\psi_{\hbar,D}(q,t) = \frac{g_{\hbar,t}(q) - g_{\hbar,t}(-q)}{\sqrt{1-\gamma_\hbar}}, \qquad \psi_{\hbar,N}(q,t) = \frac{g_{\hbar,t}(q) + g_{\hbar,t}(-q)}{\sqrt{1+\gamma_\hbar}}, \quad q > 0. \tag{3}$$

They solve the free Schrödinger equation with Dirichlet and Neumann conditions, respectively. Their regularity and decay justify the boundary calculations below.

The free Gaussian formula gives

$$|g_{\hbar,t}(0)|^2 = \sqrt{\frac{m\omega}{\pi\hbar A(t)}} \exp\left[-\frac{m\omega a(t)^2}{\hbar A(t)}\right], \qquad \hbar\,\frac{g'_{\hbar,t}(0)}{g_{\hbar,t}(0)} = D(t) := \frac{m\omega q_0 + ip_0}{1 + i\omega t}. \tag{4}$$

The simplification in D follows from $a = q_0 + p_0 t/m$; $g_{\hbar,t}(0)$ is never zero. Define $B = (m\omega q_0)^2 + p_0^2$. Since $\psi'_{\hbar,D}(0,t) = 2g'_{\hbar,t}(0)/\sqrt{1-\gamma_\hbar}$, the boundary-force density is exactly

$$F_{\hbar,D}(t) = \frac{2B}{m(1-\gamma_\hbar)A(t)} \sqrt{\frac{m\omega}{\pi\hbar A(t)}} \exp\left[-\frac{m\omega a(t)^2}{\hbar A(t)}\right]. \tag{5}$$

Position through impact and current away from impact

Write $a(t) = q_0 + p_0 t/m$, and let $g_{\hbar,t}$ be the freely evolved Gaussian. Its position variance is

$$\sigma_q(t)^2 = \frac{\hbar}{2m\omega}(1 + \omega^2 t^2). \tag{P21}$$

Because $||x| - |a|| \le |x - a|$ and reflection commutes with the free evolution,

$$\left|\langle Q\rangle_{\psi_\hbar(t)} - |a(t)|\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\, \sigma_q(t). \tag{P22}$$

Indeed the half-line position expectation equals the full-line expectation of $|X|$ in the normalized full-line odd state $\phi_\hbar = (g_\hbar - Rg_\hbar)/\sqrt{2(1-\gamma_\hbar)}$; apply the triangle inequality to the two Gaussian terms. Thus position converges uniformly on every bounded time interval, **including the impact time**, to the elastic classical folded trajectory.

Away from the impact, the corresponding probability-current momentum also converges to the incoming or reflected value. More precisely, if $|a(t)| \geq \eta > 0$, with $\sigma_p = \sqrt{\hbar m\omega/2}$,

$$\left| \int_0^\infty \overline{\psi_\hbar}(-i\hbar\partial_q)\psi_\hbar \, dq - p_0 \operatorname{sgn} a(t) \right| \leq \sqrt{\frac{2}{1-\gamma_\hbar}} \left(\sigma_p + \frac{2|p_0|\sigma_q(t)}{\eta} \right). \tag{P23}$$

For the proof, the half-line integral is the full-line expectation of $\operatorname{sgn}(X)P$ in the normalized full-line odd state $\phi_\hbar = (g_\hbar - Rg_\hbar)/\sqrt{2(1-\gamma_\hbar)}$. On the Gaussian, $\|(\operatorname{sgn} X\, P - \operatorname{sgn} a\, p_0)g_t\|$ is at most $\sigma_p + 2|p_0|\mathbb{P}(aX < 0)^{1/2}$; Chebyshev bounds the last probability by $\sigma_q(t)^2/a(t)^2$. The reflected Gaussian obeys the same estimate. For each fixed $T, \eta > 0$ independent of $\hbar$, (P23) proves uniform convergence over $|t| \leq T$ with $|a(t)| \geq \eta$. It gives no uniform bound in an $\hbar$-dependent impact layer $|a(t)| \sim \sqrt{\hbar}$ or momentum value at t_*. The expression in (P23) is a real current form on this domain, not an expectation of a self-adjoint half-line momentum.

Position converges uniformly through impact; current cannot converge uniformly to a discontinuous momentum jump. The next calculation resolves that layer in the weak measure topology.

2. The pulse and the limiting impulse

Put $A_* = A(t_*)$ and

$$\Lambda = |v|\sqrt{\frac{m\omega}{A_*}} > 0. \tag{6}$$

Because $m\omega q_0 = -p_0\omega t_*$, one has $B = p_0^2 A_*$. At every fixed $s \in \mathbb{R}$, (5) gives

$$\sqrt{\hbar}\, F_{\hbar,D}(t_* + \sqrt{\hbar}\, s) \longrightarrow \frac{2p_0^2}{m}\sqrt{\frac{m\omega}{\pi A_*}}\, e^{-\Lambda^2 s^2}. \tag{7}$$

The integral of the right side is $2|p_0|$.

This pointwise calculation proves (1) with the following tail control. For any fixed $\delta > 0$,

$$c_\delta := \inf_{|t-t_*|\geq\delta} \frac{m\omega a(t)^2}{A(t)} > 0 :$$

the expression is continuous, has only the one zero t_*, and tends to $mv^2/\omega > 0$ at both infinities. Equation (5) bounds the total force outside this neighbourhood by $C_\delta \hbar^{-1/2} e^{-c_\delta/\hbar}$, since $A(t)^{-3/2}$ is integrable. Inside the neighbourhood, after $t = t_* + \sqrt{\hbar}s$, the profiles have a common integrable Gaussian majorant. Dominated convergence proves convergence against every bounded continuous test function and gives the total mass $2|p_0|$. This is weak convergence of measures, not total-variation convergence to an atom.

The real current-momentum form is

$$J_\hbar(t) = \hbar \operatorname{Im} \int_0^\infty \overline{\psi_\hbar}\, \partial_q \psi_\hbar \, dq. \tag{8}$$

The differential operator $-i\hbar\partial_q$ on a half-line is not a self-adjoint momentum operator; (8) is a real current form. For a sufficiently regular free half-line wave,

$$\frac{dJ_\hbar}{dt} = \frac{\hbar^2}{2m}\left(|\psi_\hbar'(0,t)|^2 - \operatorname{Re}[\overline{\psi_\hbar(0,t)}\psi_\hbar''(0,t)] \right). \tag{9}$$

To see this, differentiate (8), insert $\partial_t\psi = i\hbar\psi''/(2m)$, and integrate by parts. For Dirichlet the second term vanishes, so $dJ_{\hbar,D}/dt = F_{\hbar,D}$. Using the off-impact current estimate above together with (7),

$$J_{\hbar,D}(t_* + \sqrt{\hbar}\, s) \longrightarrow |p_0| \operatorname{erf}(\Lambda s). \tag{10}$$

The incoming value is $p_0 = -|p_0|$, the outgoing value is $|p_0|$, and their difference is the mass in (1). The transition is continuous for each $\hbar > 0$ and becomes discontinuous only in the limit.

3. Neumann and Robin boundary information

For $\psi'_{\hbar,N}(0,t) = 0$, the expression in (1) is identically zero. Equation (9), rather than the Dirichlet-only term, is the appropriate boundary stress. Since

$$\frac{g''_{\hbar,t}(0)}{g_{\hbar,t}(0)} = \frac{D(t)^2}{\hbar^2} - \frac{m\omega}{\hbar(1+i\omega t)},$$

it is explicitly

$$F_{\hbar,N}(t) = \frac{2|g_{\hbar,t}(0)|^2}{m(1+\gamma_\hbar)}\left[-\operatorname{Re} D(t)^2 + \frac{\hbar m\omega}{A(t)}\right]. \tag{11}$$

This density need not be pointwise nonnegative. At the impact, $D(t_*) = ip_0$, and therefore the rescaled Neumann density has the same limit (7). Moreover

$$|F_{\hbar,N}(t)| \le \frac{2|g_{\hbar,t}(0)|^2}{m(1+\gamma_\hbar)}\left[\frac{B}{A(t)} + \frac{\hbar m\omega}{A(t)}\right],$$

so the same argument after (7) controls the total variation outside every impact neighbourhood. Consequently the finite signed measures $F_{\hbar,N}(t)dt$ converge weakly to the same $2|p_0|\delta_{t_*}$. Their impact-layer current has the same limit as (10). The even-reflection argument also gives the same folded classical position $|a(t)|$. Thus the limiting position and impulse do not recover the Dirichlet/Neumann reflection phase.

All self-adjoint realizations of the free half-line Laplacian use Dirichlet or a real Robin condition $\psi'(0) = \kappa\psi(0)$, including Neumann at $\kappa = 0$. For an incoming stationary wave $e^{-ikq} + r_\kappa(k)e^{ikq}$,

$$r_\kappa(k) = \frac{ik+\kappa}{ik-\kappa}, \qquad |r_\kappa(k)| = 1, \quad r_0 = 1, \quad r_D = -1. \tag{12}$$

At fixed real κ and fixed nonzero physical momentum, $k = |p|/\hbar \to \infty$ and $r_\kappa(k) \to 1$. This is a stationary scattering comparison, not a proof of the impulse limit for every Robin packet. For $\kappa < 0$ there is additionally a bound state $\sqrt{-2\kappa}e^{\kappa q}$ of energy $-\hbar^2\kappa^2/(2m)$, showing another quantum distinction that the free incoming bounce does not encode.

4. Consequence for STATE

For a specified coherent packet, the Dirichlet force is a positive boundary measure converging to Newton's impulsive momentum jump; the full Neumann boundary stress has the same limit. A classical bounce does not identify the boundary condition or reflection phase. This constructs neither a general boundary-groupoid theorem nor a positive-action-necessity result.